

AI ASSISTED CODING

ASSIGNMENT-13.5

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BATCH-50

Task Description #1 (Refactoring – Removing Global Variables)

- **Task:** Use AI to eliminate unnecessary global variables from the code.

- **Instructions:**

- o Identify global variables used across functions.

- o Refactor the code to pass values using function parameters.

- o Improve modularity and testability.

- **Sample Legacy Code:**

```
rate = 0.1
```

```
def calculate_interest(amount):
```

```
    return amount * rate
```

```
print(calculate_interest(1000))
```

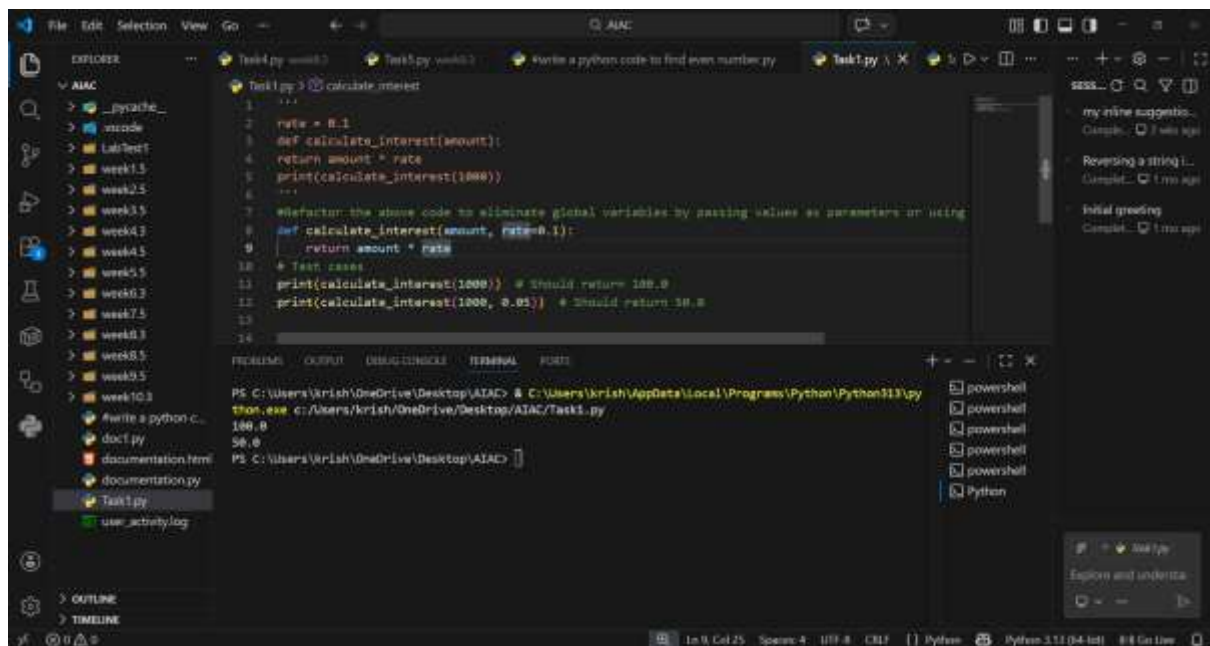
- **Expected Output:**

- o Refactored version passing rate as a parameter or using a configuration structure.

PROMPT:

“Refactor the above code to eliminate global variables by passing values as parameters or using configuration objects.”

CODE AND OUTPUT:



The screenshot shows a VS Code editor with a file explorer on the left containing a project named 'AIAC'. The main editor window displays a Python file 'Task1.py' with the following code:

```
1. """
2. rate = 0.1
3. def calculate_interest(amount):
4.     return amount * rate
5. print(calculate_interest(1000))
6. """
7. #Refactor the above code to eliminate global variables by passing values as parameters or using
8. def calculate_interest(amount, rate=0.1):
9.     return amount * rate
10. # Test cases
11. print(calculate_interest(1000)) # Should return 100.0
12. print(calculate_interest(1000, 0.05)) # Should return 50.0
13.
14.
```

The bottom panel shows the 'TERMINAL' with the following output:

```
PS C:\Users\krish\OneDrive\Desktop\AIAC> C:\Users\krish\AppData\Local\Programs\Python\Python313\py
thon.exe c:/Users/krish/OneDrive/Desktop/AIAC/Task1.py
100.0
50.0
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```

EXPLANATION:

This ensures modularity and testability by avoiding hidden dependencies. Functions become reusable since they rely only on inputs, not external state. It improves clarity and makes unit testing easier.

Task Description #2 : (Refactoring Deeply Nested Conditionals)

- **Task:** Use AI to refactor deeply nested if–elif–else logic into a cleaner structure.

- **Focus Areas:**

- o Readability

- o Logical simplification

- o Maintainability

Legacy Code:

score = 78

if score >= 90:

print("Excellent")

else:

if score >= 75:

print("Very Good")

else:

if score >= 60:

print("Good")

else:

print("Needs Improvement")

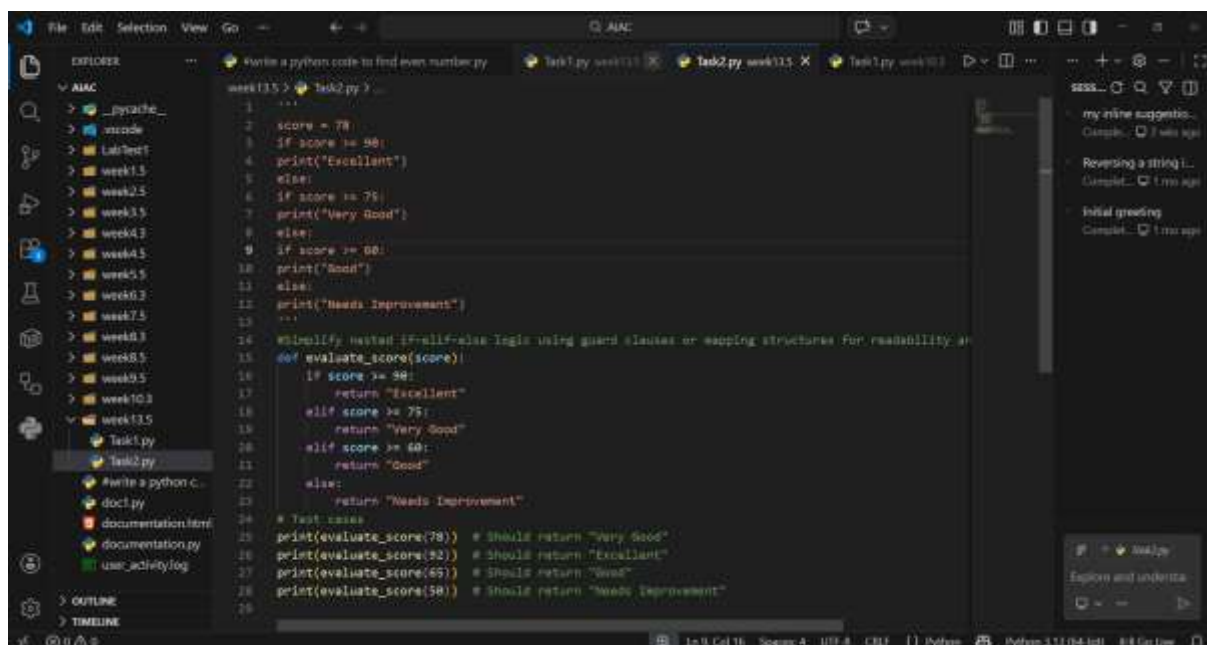
Expected Outcome:

o Flattened logic using guard clauses or a mapping-based approach.

PROMPT:

“Simplify nested if–elif–else logic using guard clauses or mapping structures for readability and maintainability.”

CODE:



```
1  """
2  score = 78
3  if score >= 90:
4      print("Excellent")
5  else:
6      if score >= 75:
7          print("Very Good")
8      else:
9          if score >= 60:
10             print("Good")
11         else:
12             print("Needs Improvement")
13     """
14  #Simplify nested if-elif-else logic using guard clauses or mapping structures for readability and maintainability.
15  def evaluate_score(score):
16      if score >= 90:
17          return "Excellent"
18      elif score >= 75:
19          return "Very Good"
20      elif score >= 60:
21          return "Good"
22      else:
23          return "Needs Improvement"
24
25  # Test cases
26  print(evaluate_score(78)) # Should return "Very Good"
27  print(evaluate_score(92)) # Should return "Excellent"
28  print(evaluate_score(65)) # Should return "Good"
29  print(evaluate_score(50)) # Should return "Needs Improvement"
```

OUTPUT:

```
thon.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task2.py
Very Good
Excellent
Good
Needs Improvement
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```

EXPLANATION:

Flattening conditionals reduces cognitive load and makes logic easier to follow. Guard clauses exit early, while mappings provide a clean lookup. This improves readability and future scalability.

Task 3 (Refactoring Repeated File Handling Code)

- **Task:** Use AI to refactor repeated file open/read/close logic.

- **Focus Areas:**

- o DRY principle

- o Context managers

- o Function reuse

Legacy Code:

```
f = open("data1.txt")
```

```
print(f.read())
```

```
f.close()
```

```
f = open("data2.txt")
```

```
print(f.read())
```

```
f.close()
```

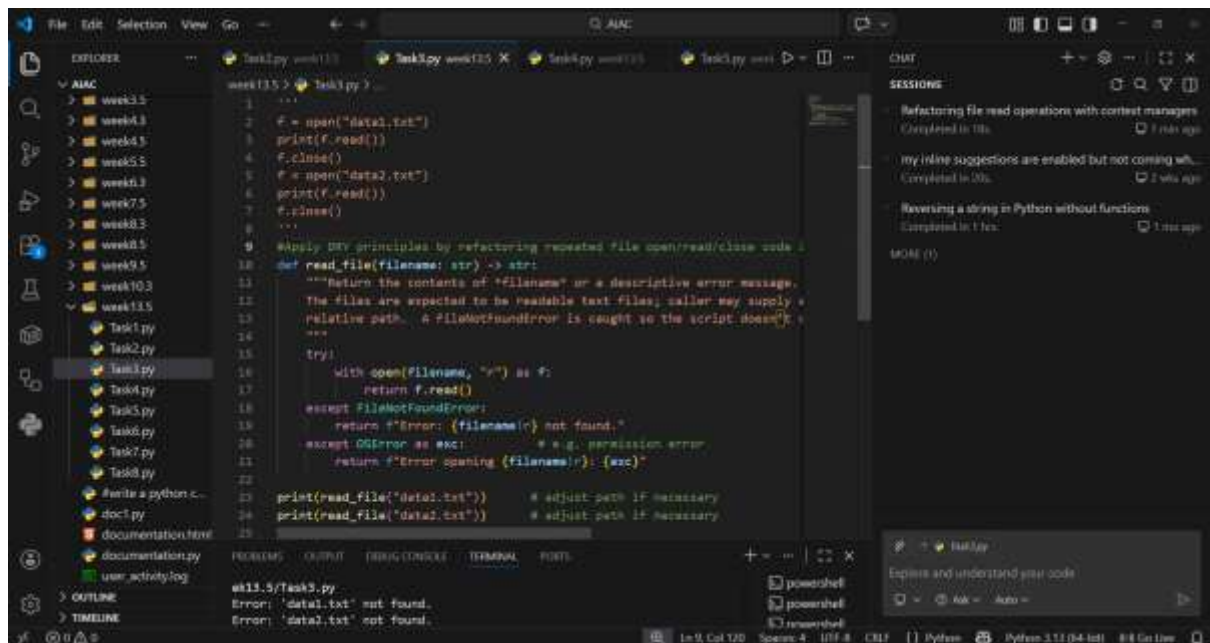
Expected Outcome:

- o Reusable function using with open() and parameters.

PROMPT:

“Apply DRY principles by refactoring repeated file open/read/close code into a reusable function with context managers.”

CODE AND OUTPUT:



```
1  ...
2  f = open("data1.txt")
3  print(f.read())
4  f.close()
5  f = open("data2.txt")
6  print(f.read())
7  f.close()
8  ...
9  Apply DRY principles by refactoring repeated file open/read/close code
10 def read_file(filename: str) -> str:
11     """Return the contents of *filename* or a descriptive error message.
12     The files are expected to be readable text files; caller may supply a
13     relative path. A FileNotFoundError is caught so the script doesn't
14     ...
15     try:
16         with open(filename, "r") as f:
17             return f.read()
18     except FileNotFoundError:
19         return f"Error: {filename} not found."
20     except OSError as exc:
21         # e.g. permission error
22         return f"Error opening {filename}: {exc}"
23
24 print(read_file("data1.txt")) # adjust path if necessary
25 print(read_file("data2.txt")) # adjust path if necessary
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

task13.5/Task3.py
Error: 'data1.txt' not found.
Error: 'data2.txt' not found.

EXPLANATION:

Using `with open()` ensures safe resource handling and avoids manual closing. Encapsulating file logic in a function reduces duplication. This makes the code concise, reusable, and less error-prone.

Task 4 (Optimizing Search Logic)

- Task: Refactor inefficient linear searches using appropriate data structures.

• Focus Areas:

o Time complexity

o Data structure choice

Legacy Code:

```
users = ["admin", "guest", "editor", "viewer"]
```

```
name = input("Enter username: ")
```

```
found = False
```

```
for u in users:
```

```
if u == name:
```

```
found = True
```

```
print("Access Granted" if found else "Access Denied")
```

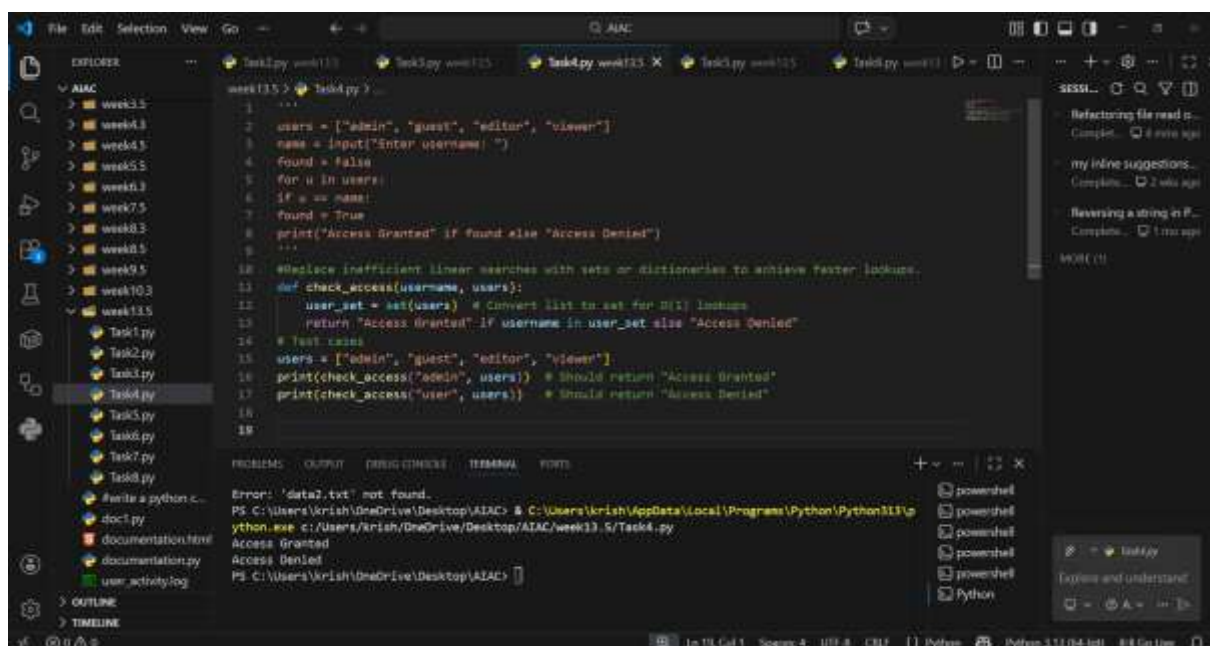
Expected Outcome:

- o Use of sets or dictionaries with complexity justification.

PROMPT:

"Replace inefficient linear searches with sets or dictionaries to achieve faster lookups."

CODE AND OUTPUT:

The screenshot shows a Visual Studio Code editor with a Python file named 'Task4.py'. The code defines a list of users and a function 'check_access' that uses a set for O(1) lookups. The output console shows the results of two function calls: 'Access Granted' for 'admin' and 'Access Denied' for 'user'.

```
1 # ...
2 users = ["admin", "guest", "editor", "viewer"]
3 name = input("Enter username: ")
4 found = False
5 for u in users:
6     if u == name:
7         found = True
8         print("Access Granted" if found else "Access Denied")
9
10 # Replace inefficient linear searches with sets or dictionaries to achieve faster lookups.
11 def check_access(username, users):
12     user_set = set(users) # Converts list to set for O(1) lookups
13     return "Access Granted" if username in user_set else "Access Denied"
14
15 # Test cases
16 users = ["admin", "guest", "editor", "viewer"]
17 print(check_access("admin", users)) # Should return "Access Granted"
18 print(check_access("user", users)) # Should return "Access Denied"
19
```

Output:

```
PS C:\Users\krish\OneDrive\Desktop\AIAC> & C:\Users\krish\AppData\Local\Programs\Python\Python311\python.exe c:\Users\krish\OneDrive\Desktop\AIAC\week33.5\Task4.py
Access Granted
Access Denied
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```

EXPLANATION:

Linear search has $O(n)$ complexity, while sets/dictionaries provide average $O(1)$ lookups. This drastically improves performance for large datasets. It also makes the code cleaner and more efficient.

Task 5 (Refactoring Procedural Code into OOP Design)

- Task: Use AI to refactor procedural code into a class-based design.

- Focus Areas:

- o Object-Oriented principles

- o Encapsulation

Legacy Code:

```
salary = 50000
```

```
tax = salary * 0.2
```

```
net = salary - tax
```

```
print(net)
```

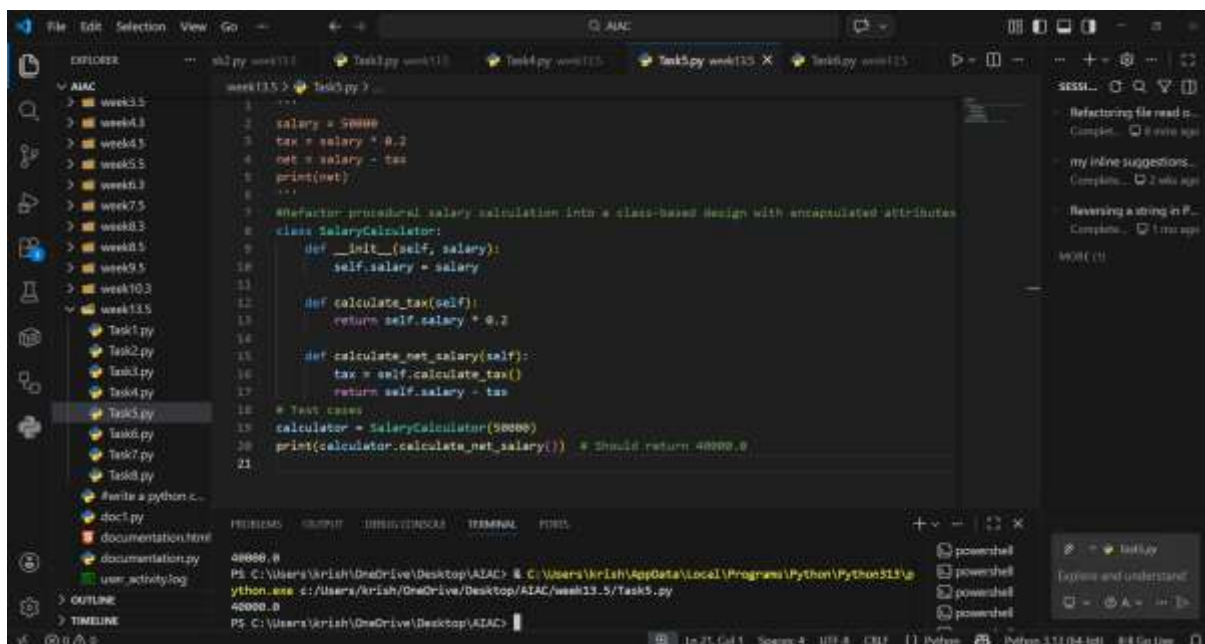
Expected Outcome:

o A class like `EmployeeSalaryCalculator` with methods and attributes.

PROMPT:

“Refactor procedural salary calculation into a class-based design with encapsulated attributes and methods.”

CODE AND OUTPUT:



```
1 salary = 50000
2 tax = salary * 0.2
3 net = salary - tax
4 print(net)
5
6 # Refactor procedural salary calculation into a class-based design with encapsulated attributes
7 class SalaryCalculator:
8     def __init__(self, salary):
9         self.salary = salary
10
11     def calculate_tax(self):
12         return self.salary * 0.2
13
14     def calculate_net_salary(self):
15         tax = self.calculate_tax()
16         return self.salary - tax
17
18 # Test cases
19 calculator = SalaryCalculator(50000)
20 print(calculator.calculate_net_salary()) # Should return 40000.0
21
```

Terminal Output:

```
40000.0
PS C:\Users\krish\OneDrive\Desktop\AIAC> E:\C:\Users\krish\AppData\Local\Programs\Python\Python311\python.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task5.py
40000.0
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```

EXPLANATION:

Object-Oriented design improves structure and reusability. Encapsulation keeps salary, tax, and net logic within a class. This makes the code extensible, testable, and aligned with OOP principles.

Task 6 (Refactoring for Performance Optimization)

- Task: Use AI to refactor a performance-heavy loop handling

large data.

- Focus Areas:

- o Algorithmic optimization

- o Use of built-in functions

Legacy Code:

```
total = 0
```

```
for i in range(1, 1000000):
```

```
    if i % 2 == 0:
```

```
        total += i
```

```
print(total)
```

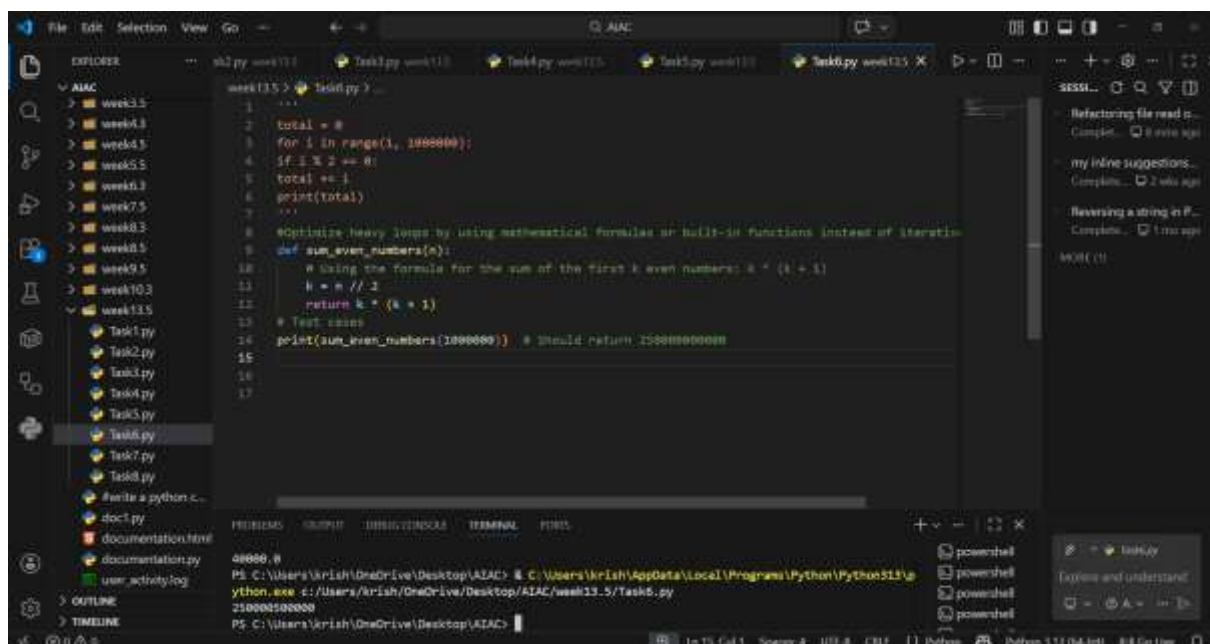
Expected Outcome:

- o Optimized logic using mathematical formulas or comprehensions, with time comparison.

PROMPT:

"Optimize heavy loops by using mathematical formulas or built-in functions instead of iterative summation."

CODE AND OUTPUT:



The screenshot shows a Visual Studio Code editor with a Python file named 'Task6.py' open. The code in the file is as follows:

```
1  """
2  total = 0
3  for i in range(1, 1000000):
4      if i % 2 == 0:
5          total += i
6      print(total)
7  """
8  #Optimize heavy loops by using mathematical formulas or built-in functions instead of iteration
9  def sum_even_numbers(n):
10     # Using the formula for the sum of the first k even numbers: k * (k + 1)
11     k = n // 2
12     return k * (k + 1)
13     # Test cases
14     print(sum_even_numbers(1000000)) # Should return 250000000000
15
16
17
```

The output of the script is displayed in the terminal at the bottom of the editor:

```
40000, 0
PS C:\Users\krish\OneDrive\Desktop\AIAC> & C:\Users\krish\AppData\Local\Programs\Python\Python311\python.exe c:\Users\krish\OneDrive\Desktop\AIAC\week13.5\Task6.py
250000000000
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```


EXPLANATION:

Loops over large ranges are slow; formulas like arithmetic progression or built-ins like `sum()` with comprehensions are faster. This reduces runtime complexity and improves efficiency significantly.

Task 7 (Removing Hidden Side Effects)

- **Task:** Refactor code that modifies shared mutable state.

- **Focus Areas:**

- o Functional-style refactoring

- o Predictability

Legacy Code:

```
data = []
```

```
def add_item(x):
```

```
    data.append(x)
```

```
add_item(10)
```

```
add_item(20)
```

```
print(data)
```

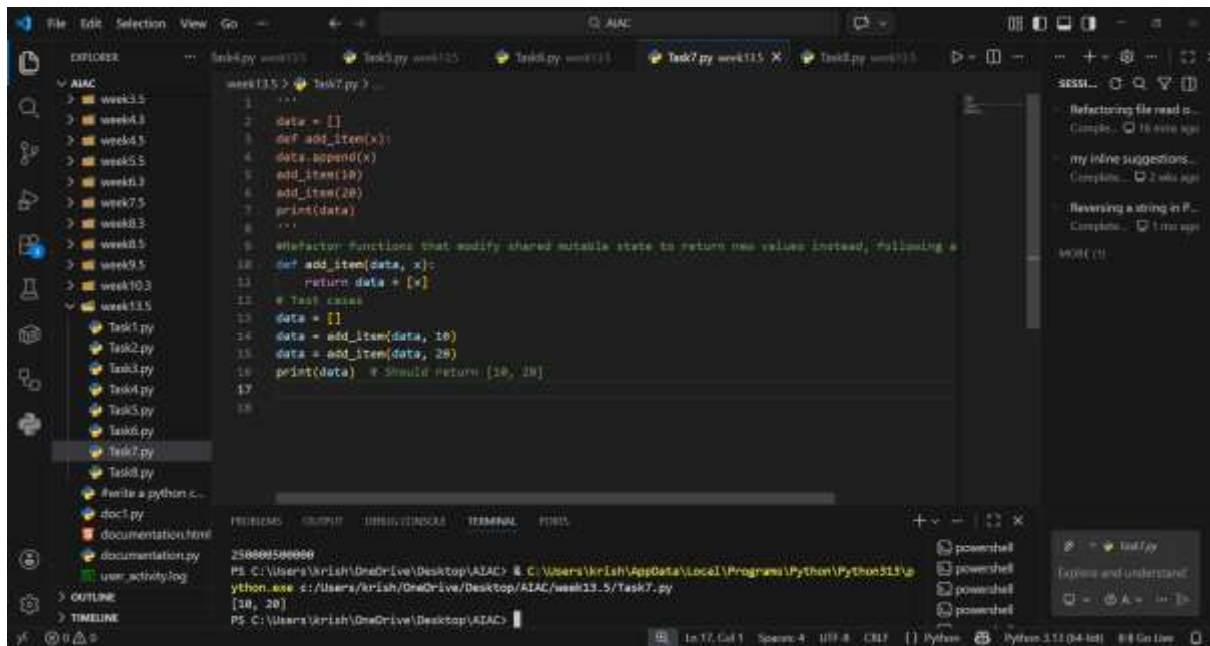
Expected Outcome:

- o Refactored function returning values instead of mutating globals.

PROMPT:

“Refactor functions that modify shared mutable state to return new values instead, following a functional style.”

CODE AND OUTPUT:



```
1  ...
2  data = []
3  def add_item(x):
4      data.append(x)
5      add_item(10)
6      add_item(20)
7      print(data)
8  ...
9  # Refactor functions that modify shared mutable state to return new values instead, following *
10 def add_item(data, x):
11     return data + [x]
12 # Test cases
13 data = []
14 data = add_item(data, 10)
15 data = add_item(data, 20)
16 print(data)  # Should return [10, 20]
17
18
```

EXPLANATION:

This avoids hidden side effects by eliminating reliance on global or shared mutable data. Returning values makes behavior predictable and easier to test. It improves modularity and ensures functions remain pure, enhancing reliability.

Task 8 (Refactoring Complex Input Validation Logic)

- **Task:** Use AI to simplify and modularize complex validation rules.

- **Focus Areas:**

- o Readability

- o Testability

Legacy Code:

```
password = input("Enter password: ")
```

```
if len(password) >= 8:
```

```
if any(c.isdigit() for c in password):
```

```
if any(c.isupper() for c in password):
```

```
print("Valid Password")
```

else:

print("Must contain uppercase")

else:

print("Must contain digit")

else:

print("Password too short")

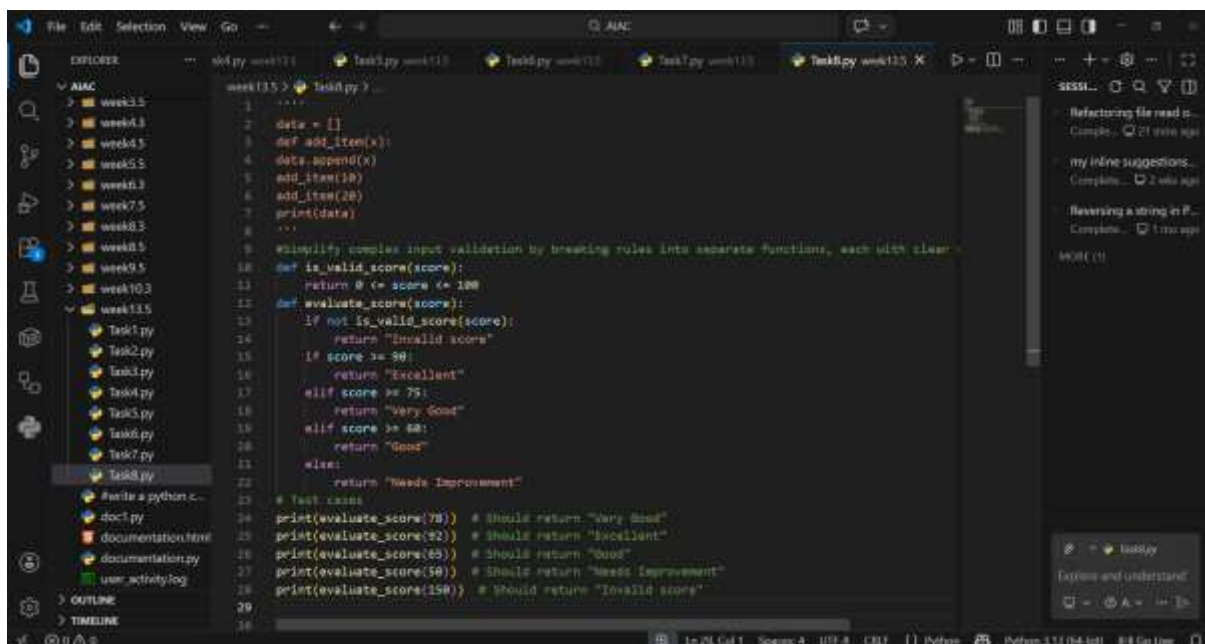
Expected Outcome:

o Separate validation functions with clear responsibility

PROMPT:

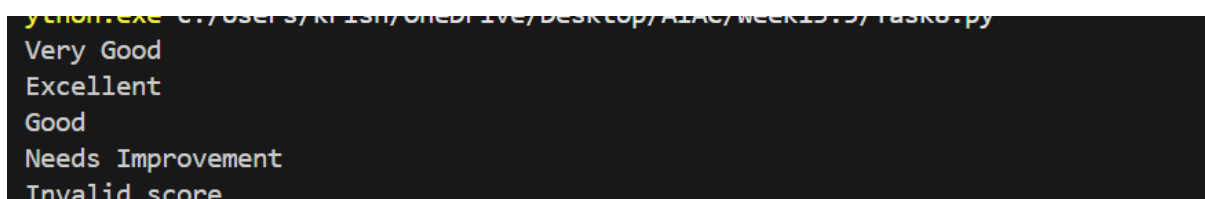
"Simplify complex input validation by breaking rules into separate functions, each with clear responsibility."

CODE:



```
1  """
2  data = []
3  def add_item(x):
4      data.append(x)
5      add_item(10)
6      add_item(20)
7      print(data)
8  """
9  #Simplify complex input validation by breaking rules into separate functions, each with clear
10
11  def is_valid_score(score):
12      return 0 <= score <= 100
13
14  def evaluate_score(score):
15      if not is_valid_score(score):
16          return "Invalid score"
17      if score >= 90:
18          return "Excellent"
19      elif score >= 75:
20          return "Very Good"
21      elif score >= 60:
22          return "Good"
23      else:
24          return "Needs Improvement"
25
26  # Test cases
27  print(evaluate_score(78)) # Should return "Very Good"
28  print(evaluate_score(92)) # Should return "Excellent"
29  print(evaluate_score(65)) # Should return "Good"
30  print(evaluate_score(50)) # Should return "Needs Improvement"
31  print(evaluate_score(150)) # Should return "Invalid score"
```

OUTPUT:



```
python.exe C:/Users/Krish/OneDrive/Desktop/AIAC/week13.5/task8.py
Very Good
Excellent
Good
Needs Improvement
Invalid score
```

EXPLANATION:

Splitting validation into modular functions improves readability and testability. Each function handles one rule (length, digit, uppercase), making the logic easier to maintain. This approach supports unit testing and future extensions without nested conditionals.